

Simulation and Modeling

CSIT — 5th Semester — 2078 — Answer Sheet
Subject Code: CSC328

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define queuing system. Explain the Kendall's notation for queuing system. What are the various performance measures in single server queuing system? Explain which of them determine system stability and how?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define true random numbers and pseudo random numbers with its properties. The sequence of numbers 0.64, 0.25, 0.58, 0.72, 0.90 has been generated. Use KS Test with $D\alpha=0.050 \Rightarrow 0.512$ to determine if the hypothesis that they are uniformly distributed on interval [0,1] can be rejected.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What do you understand by dynamic mathematical model? Explain with example. Differentiate it with static mathematical model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Describe the phases in simulation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Explain the concept of discrete event simulation. Explain poisson's arrival pattern.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain Monte Carlo simulation method with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define the terms verification, calibration, validation and accreditation of models

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Use multiplicative congruential method to generate a sequence of random numbers with $x_0 = 7$, $a = 11$, $m = 16$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Why is estimation methods used in simulation? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the importance of elimination of initial bias during simulation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Workers come to a supply store at the rate of one every 6 ± 2 minute. Their requisitions are processed by one of the two clerks who take 8 ± 2 minutes for each requisition. The requisitions are then passed to a single storekeeper who fills them one at a time, taking 6 ± 3 minutes for each. Draw GPSS block diagram to simulate the above problem for 100 requisitions.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on (any two): a. Digital analog simulator b. Simulation tools

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.